

AL-AZHARI LIBRARY — Backend Phase 5 Implementation Report

1. Status

Phase 5 Complete — 100% Verified

- TypeScript Typecheck: **PASS** (0 errors)
- ESLint: **PASS** (0 errors , 0 warnings)
- Production Build: **PASS** (tsc emits cleanly to dist/)
- Automated Tests: **PASS** (49 test suites , 385 tests passing , 0 failures)
- Development Server: **PASS** (npm run dev)
- Health Probes: /health/live **PASS** (200 OK), /health/ready **PASS** (200 OK, database connected)
- Public Catalog Smoke Tests: /api/v1/products , /api/v1/categories , /api/v1/search , /api/v1/content/home **PASS**

2. Files Created

- src/common/types/localized.ts — LocalizedText interface ({ ar: string; en?: string })
- src/modules/categories/utils/category.projection.ts — Safe public category projection (toSafeCategory)
- src/modules/products/utils/search-normalizer.ts — Resilient search normalization for Arabic diacritics/variants and Latin text (buildSearchText , normalizeText , escapeRegex)
- src/modules/products/utils/product.projection.ts — Public and admin product/variant projections hiding stock counters
- src/database/migrations/scripts/20260927_003_catalog.migration.ts — Non-destructive idempotent index migration
- docs/PHASE-5-CATALOG.md — Authoritative Phase 5 documentation
- tests/unit/category-schema.test.ts — Unit tests for category validation schemas
- tests/unit/product-schema.test.ts — Unit tests for product/variant validation schemas and money rules
- tests/unit/search-normalizer.test.ts — Unit tests for Arabic tashkeel removal, alef normalization, and ReDoS prevention
- tests/unit/catalog-migration.test.ts — Unit tests for idempotent up/down migration
- tests/api/category-api.test.ts — Integration tests for public and admin category APIs
- tests/api/product-api.test.ts — Integration tests for public catalog and admin product management
- tests/api/search-api.test.ts — Integration tests for public search (Arabic/Latin, injection defense, bounds)
- tests/api/content-api.test.ts — Integration tests for public home content and admin content modules
- tests/api/catalog-security.test.ts — Integration tests for RBAC permissions, mass-assignment blocking, and privacy checks
- tests/integration/variant.test.ts — Integration tests for product variants, pricing rules, and safe public projection

3. Files Modified

- src/common/types/index.ts — Exported LocalizedText
- src/common/validators/common.validators.ts — Added localizedTextSchema and localizedDescriptionSchema
- src/modules/users/services/roles.service.ts — Added content.write to Admin role definition
- src/modules/audit/types/audit.types.ts — Minimal audit type definitions
- src/modules/audit/models/audit-log.model.ts — AuditLogModel on collection auditLogs
- src/modules/audit/services/audit.service.ts — Safe, non-blocking auditService.record()

7. `src/modules/audit/index.ts` — Audit module exports
8. `src/modules/categories/types/category.types.ts` — Category domain interfaces and DTOs
9. `src/modules/categories/models/category.model.ts` — `CategoryModel` with books-first index
10. `src/modules/categories/repositories/category.repository.ts` — Category database repository
11. `src/modules/categories/services/category.service.ts` — Category service with deactivation and historical dependency protection
12. `src/modules/categories/controllers/category.controller.ts` — Category public and admin HTTP controllers
13. `src/modules/categories/schemas/category.schema.ts` — Zod strict validation schemas
14. `src/modules/categories/routes/category.routes.ts` — Public `/categories` and Admin `/admin/categories` routers
15. `src/modules/categories/index.ts` — Categories module exports
16. `src/modules/products/types/product.types.ts` — Product and variant domain interfaces, projections, and DTOs
17. `src/modules/products/models/product.model.ts` — `ProductModel` with embedded variants and strict money validators
18. `src/modules/products/repositories/product.repository.ts` — Product queries for public browsing, search, and admin
19. `src/modules/products/services/product.service.ts` — Product service with publication constraints and audit hooks
20. `src/modules/products/controllers/product.controller.ts` — Product public, search, and admin HTTP controllers
21. `src/modules/products/schemas/product.schema.ts` — Zod strict validation schemas for products, variants, and filters
22. `src/modules/products/routes/product.routes.ts` — Public `/products`, `/search`, and Admin `/admin/products` routers
23. `src/modules/products/index.ts` — Products module exports
24. `src/modules/content/types/content.types.ts` — Content module domain interfaces and DTOs
25. `src/modules/content/models/content-module.model.ts` — `ContentModuleModel` on collection `contentModules`
26. `src/modules/content/repositories/content.repository.ts` — Date-window and admin content queries
27. `src/modules/content/services/content.service.ts` — Content service resolving referenced published products and active categories
28. `src/modules/content/controllers/content.controller.ts` — Content public `/content/home` and admin HTTP controllers
29. `src/modules/content/schemas/content.schema.ts` — Content module Zod schemas and date range refinements
30. `src/modules/content/routes/content.routes.ts` — Content routers
31. `src/modules/content/index.ts` — Content module exports
32. `src/database/migrations/index.ts` — Registered `catalogMigration`
33. `src/app.ts` — Mounted public `/categories`, `/products`, `/search`, `/content` and `/admin/*` routes
34. `README.md` — Updated documentation index with Phase 5 link

4. Product Implementation Summary

- **Schema & Model:** `ProductModel` on `products` collection with `strict: 'throw', timestamps: true`.
- **Money Representation:** Strictly integer `priceMinor` (piastres, EGP). Verified non-negative with integer validators. Floating-point values rejected.
- **Descriptions:** Nullable per authoritative PRD specification (`description === null` is valid and does NOT block publication).
- **Availability States:** Enforced to `'in_stock' | 'out_of_stock' | 'pre_order_eligible'`.
- **Publication Rules:** Products can only be published if they have valid pricing, a valid category reference, and the category is active and MVP-enabled.
- **Audit Integration:** Critical mutations (price change, availability transition, publication toggle) create audit entries via `auditService.record()`.

5. Category Implementation Summary

- **Schema & Model:** CategoryModel on `categories` collection with `strict: 'throw', timestamps: true`.
- **Books-First Prioritization:** Public category listing orders by `{ isBooksCore: -1, displayOrder: 1, createdAt: 1 }`.
- **MVP Enablement Filtering:** Public category listing strictly serves `isActive: true, isMvpEnabled: true`.
- **Non-Destructive Deactivation:** Normal category deletion sets `isActive: false`.
- **Historical Dependency Protection:** If products or child subcategories reference the category, deletion is prevented with `409 RESOURCE_CONFLICT`.

6. Variant Implementation Summary

- **Embedded Subdocuments:** Embedded inside `ProductModel.variants` with `_id: false`.
- **Identity & Labels:** Stable unique `variantId` per product, localized Arabic/English `label`.
- **Pricing & Availability:** Independent `priceMinor`, currency ('EGP'), `availability`, and `preOrderEligible`.
- **Internal Counters Shielding:** Variant `stockTotal` and `stockReserved` are strictly stripped from all public projections.

7. Content Implementation Summary

- **Schema & Model:** ContentModuleModel on `contentModules` collection.
- **Module Types:** Supported types include `'hero_banner', 'featured_products', 'category_grid', 'announcement', 'promo_banner', 'text_block'`.
- **Date Windows:** Public `/content/home` queries filter `active: true` where `now` is between `startsAt` and `endsAt`.
- **Entity Protection:** Referenced unpublished products or inactive categories are pruned from the public response payload.

8. Search Implementation Summary

- **Normalization Engine:** `search-normalizer`ts normalizes Arabic tashkeel (diacritics), alef variants (ا , آ , إ , ؤ), teh marbuta (ة), alef maksura (ي), and tatweel (ـ).
- **Latin Queries:** Case-insensitive and trimmed.
- **Security:** User query regex characters are escaped via `escapeRegex()` to block ReDoS and NoSQL injection.
- **Bounds:** Strict pagination (`maxLimit: 100`) and input validation.

9. API Endpoints

Public

- `GET /api/v1/categories` — Public active + MVP-enabled categories in books-first order
- `GET /api/v1/products` — Bounded published product catalog listing
- `GET /api/v1/products/:slug` — Published product detail lookup
- `GET /api/v1/search` — Normalized catalog search
- `GET /api/v1/content/home` — Dynamic homepage merchandising modules

Admin

- GET /api/v1/admin/categories — List all categories (categories.write)
- POST /api/v1/admin/categories — Create category (categories.write)
- PATCH /api/v1/admin/categories/:id — Update category (categories.write)
- DELETE /api/v1/admin/categories/:id — Deactivate category (categories.write)
- GET /api/v1/admin/products — List all products with stock counters (products.write)
- GET /api/v1/admin/products/:id — Product detail with internal counters (products.write)
- POST /api/v1/admin/products — Create product (products.write)
- PATCH /api/v1/admin/products/:id — Update product (products.write)
- GET /api/v1/admin/content — List content modules (content.write)
- GET /api/v1/admin/content/:id — Content module detail (content.write)
- POST /api/v1/admin/content — Create content module (content.write)
- PATCH /api/v1/admin/content/:id — Update content module (content.write)
- DELETE /api/v1/admin/content/:id — Delete content module (content.write)

10. Database / Index / Migration Changes

- **Migration Script:** src/database/migrations/scripts/20260927_003_catalog.migration.ts
- **Categories Indexes:**
 - idx_categories_slug_unique : { slug: 1 } (unique)
 - idx_categories_public_browse : { isActive: 1, isMvpEnabled: 1, isBooksCore: -1, displayOrder: 1 }
- **Products Indexes:**
 - idx_products_slug_unique : { slug: 1 } (unique)
 - idx_products_category_published_display : { categoryId: 1, isPublished: 1, displayOrder: 1 }
 - idx_products_availability_published : { availability: 1, isPublished: 1 }
 - idx_products_published_search : { isPublished: 1, searchText: 1 }
 - idx_products_isbn_sparse_unique : { 'metadata.isbn': 1 } (unique, partialFilterExpression: { 'metadata.isbn': { \$type: 'string' } })
- **Content Modules Indexes:**
 - idx_content_modules_key_unique : { key: 1 } (unique)
 - idx_content_modules_active_display : { active: 1, displayOrder: 1, startsAt: 1, endsAt: 1 }
- **Audit Logs Indexes:**
 - idx_audit_logs_entity : { entityType: 1, entityId: 1, createdAt: -1 }
 - idx_audit_logs_actor : { actorId: 1, createdAt: -1 }

11. Security Controls

- **RBAC:** Requires authenticated principals with explicit permissions (products.write , categories.write , content.write). Store Owner retains universal authority.
- **Mass-Assignment Defense:** Strict Zod schemas (.strict()) reject arbitrary field injections (_id , role , status , createdAt , etc.).
- **Public Information Hiding:** Public responses project safe DTOs omitting stockTotal , stockReserved , isPublished , searchText , and internal operational metadata.
- **NoSQL Injection Defense:** User queries are strongly typed and validated against allowlists; raw Mongo operators (\$where , \$regex , \$ne) are blocked.

12. Tests Created

- tests/unit/category-schema.test.ts (7 tests)
- tests/unit/product-schema.test.ts (17 tests)
- tests/unit/search-normalizer.test.ts (8 tests)
- tests/unit/catalog-migration.test.ts (2 tests)
- tests/integration/variant.test.ts (4 tests)
- tests/api/category-api.test.ts (8 tests)
- tests/api/product-api.test.ts (13 tests)
- tests/api/search-api.test.ts (9 tests)
- tests/api/content-api.test.ts (7 tests)
- tests/api/catalog-security.test.ts (6 tests)

13. Total Tests Passed

- **Phase 4 Baseline:** 39 test suites, 297 tests
- **Phase 5 Total:** 49 test suites, 385 tests passed (100% pass rate, 0 failures)

14. Verification Commands & Results

```
npm run typecheck    # PASS (0 errors)
npm run lint         # PASS (0 errors, 0 warnings)
npm run build        # PASS (tsc compiled dist/)
npm test             # PASS (49 suites, 385 tests)
npm run dev          # PASS (server runs on port 3000)
curl -s http://localhost:3000/health/live    # {"success":true,"data":{"status":"ok"}}
curl -s http://localhost:3000/health/ready   # {"success":true,"data":{"status":"ready","database":"connected"}}
```

15. Database Safety Confirmation

- No collections were dropped (`drop()` , `dropDatabase()` never invoked).
- No destructive deletions (`deleteMany()` , `remove()` against production DB never invoked).
- All tests ran hermetically on ephemeral `mongodb-memory-server` instances.
- Real database state and existing test-independent collections remain intact.

16. Deferred Items

- **Phase 6:** Cart module
- **Phase 7:** Inventory reservation ledger and deduction workflows
- **Phase 8:** Checkout engine
- **Phase 9:** Payments & payment proof processing
- **Phase 10:** Shipping & Coupon engines
- **Later Phases:** Pre-orders workflow, Customer Reviews & Ratings, Reports & BI

17. Regression Status

- **Zero Regressions:** All 39 test suites from Phases 0, 1, 2, 3, and 4 continue to pass with 100% success rate alongside the 10 new Phase 5 suites.

18. Confirmation of Phase Boundary

No Phase 6 work, Cart schemas, Cart endpoints, or Cart models were implemented. Phase 5 work is strictly isolated and complete.

Phase 5 complete. Implementation stopped here. No Phase 6 work was implemented.